Year 12 Further Maths - Matrices

Similar questions - Transformations, invariant lines and products

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The 2×2 matrix **A** represents a transformation T with the following properties.

- The image of the point $(1,0)$ is the point $(4,3)$.
- An object shape of area 5 is transformed to an image shape of area 15.
- T has a line of invariant points.

(a) Find a possible matrix for **A**. **[7]**

(b) The transformation S is represented by $\mathbf{B} = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$. Find the equation of the line of invariant points of S. **[2]**

(c) Show that any line of the form $y = x + c$ is an invariant line of S. **[3]**

2 The matrices **A** and **B** are given by $\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}$.

(a) Use **A** and **B** to disprove the proposition "matrix multiplication is commutative". **[2]**

(b) Matrix **B** represents the transformation T_B. Describe T_B fully. **[2]**

(c) By considering the inverse transformation, determine $\mathbf{B}^{-1}$. **[2]**

(d) Matrix $\mathbf{C} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$ represents T_C, and T_{BC} is T_C followed by T_B. An object of area 5 is transformed by T_{BC}. Find the area of the image. **[2]**

3 The matrix $\mathbf{R} = \begin{pmatrix} a & 0 & b \\ 0 & 1 & 0 \\ -b & 0 & a \end{pmatrix}$, where $a = \frac{\sqrt{3}}{2}$ and $b = \frac{1}{2}$.

(a) Find $\mathbf{R}^2$ in terms of a and b, and hence exactly. **[4]**

(b) Find $\mathbf{R}^6$, $\mathbf{R}^{12}$ and $\mathbf{R}^{24}$. **[3]**

(c) Describe fully the transformation represented by **R**. **[3]**

4 The matrices $\mathbf{P} = \begin{pmatrix} 1 & k & 0 \\ -2 & 1 & 3 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} 1+k \\ -1 \\ 2 \end{pmatrix}$, where k is a constant. Exactly one of the statements is true.

Statement A: **P** and **Q** (in that order) are conformable for multiplication. Statement B: **Q** and **P** (in that order) are conformable for multiplication.

(a) State, with a reason, which statement is true. **[2]**

(b) Find the product in terms of k. **[2]**

Question 1

(a) $\mathbf{A} = \begin{pmatrix} 4 & b \\ 3 & d \end{pmatrix}$ with $\det\mathbf{A} = \pm 3$ (area factor 3) and eigenvalue 1 (line of invariant points): $\det(\mathbf{A}-\mathbf{I}) = 3(d-1)-3b = 0$, so $d = b+1$. Taking $\det = 3$: $4d-3b = 3 \Rightarrow b = -1$, $d = 0$, giving $\mathbf{A} = \begin{pmatrix} 4 & -1 \\ 3 & 0 \end{pmatrix}$ (another possibility with $\det = -3$ is $\begin{pmatrix} 4 & -7 \\ 3 & -6 \end{pmatrix}$)

(b) $(\mathbf{B}-\mathbf{I})\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0}$: $2x+y = 0$, so the line is $y = -2x$

(c) $\mathbf{B}\begin{pmatrix} x \\ x+c \end{pmatrix} = \begin{pmatrix} 4x+c \\ 4x+2c \end{pmatrix}$ and $(4x+2c)-(4x+c) = c$, so the image of any point on $y = x+c$ lies on $y = x+c$

Question 2

(a) $\mathbf{AB} = \begin{pmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}$ but $\mathbf{BA} = \begin{pmatrix} -\frac{3}{5} & -\frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{pmatrix}$, so $\mathbf{AB} \neq \mathbf{BA}$

(b) A rotation about the origin through $\tan^{-1}\frac{4}{3} = 53.1^\circ$ anticlockwise

(c) The inverse is a rotation of 53.1° clockwise: $\mathbf{B}^{-1} = \begin{pmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{pmatrix}$

(d) $\det\mathbf{BC} = 1 \times (-2) = -2$, so the area is $5 \times 2 = 10$

Question 3

(a) $\mathbf{R}^2 = \begin{pmatrix} a^2-b^2 & 0 & 2ab \\ 0 & 1 & 0 \\ -2ab & 0 & a^2-b^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \end{pmatrix}$

(b) $\mathbf{R}^6 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ (rotation of 180°), $\mathbf{R}^{12} = \mathbf{I}$, $\mathbf{R}^{24} = \mathbf{I}$

(c) A rotation of 30° about the y-axis (the x-axis is mapped towards the negative z-axis)

Question 4

(a) $\mathbf{P}$ is 2×3 and $\mathbf{Q}$ is 3×1: the number of columns of $\mathbf{P}$ equals the number of rows of $\mathbf{Q}$, so Statement A is true (for $\mathbf{QP}$ the columns of $\mathbf{Q}$ (1) would need to equal the rows of $\mathbf{P}$ (2))

(b) $\mathbf{PQ} = \begin{pmatrix} (1+k)-k+0 \\ -2(1+k)-1+6 \end{pmatrix} = \begin{pmatrix} 1 \\ 3-2k \end{pmatrix}$